Label review packet: 15 items, one per page

2 November 2026. For the reviewer and an independent second checker. Each page shows one item from `label_review.TEMPLATE.json`:

- what you confirm;
- the current label;
- the wording of the original documents, quoted with the page;
- what our data show (a cross-check only, never the source of a label);
- the assumptions;
- the recommendation.

Nothing in this packet is approved. Every `confirmed` field is empty.

Under review: `results/supervised/label_table.json`, SHA-256 4eff40d85f35e35d… (labels unchanged; wording of items 2 and 15 revised 2 November)

Original documents (open them at the cited page):

Short name	Document	Where	SHA-256 (first 16)
Description	O. K. Aimiyekagbon, "Run-to-failure data set of ball bearings subjected to time-varying load and speed conditions", v2, 25 March 2024, 9 pages (CC BY 4.0)	Zenodo 10.5281/zenodo.10805042, file `LDM_UPB_Run-to-failure_data_set_of_ball_bearings.pdf`	949c855fa0cc6d2b
B01 log	Experiment B01 log, 2 pages	`model/newdata/B01/B01_log.pdf`	4a5445c8ac9f9249
B02 log	Experiment B02 log, 2 pages	`model/newdata/B02/B02_log.pdf`	5beffbeebd5ffa1f
B03 log	Experiment B03 log, 2 pages	`model/newdata/B03/B03_log.pdf`	caa9c543658ff374
B04 log	Experiment B04 log, 2 pages	`model/newdata/B04/B04_log.pdf`	84ea2506dbeaaa14

All four logs were written by Lennart Viemann and supervised by O. K. Aimiyekagbon (log p.2).

How to use this packet:

1. **Reviewer:** read each page against the original document. Tick confirm, or change, and add a comment.
2. **Second checker:** do the same independently, without seeing the reviewer's ticks.
3. **Disagreements:** settle them before sign-off. If any label changes, I regenerate the table; its hash changes and this packet is reissued.
4. **Sign-off:** record it in `results/supervised/label_review.json`, a copy of the template with every `confirmed` set to true. The reviewer fills in `checked_by` and `date`; the second checker fills in `independent_check_by` and `independent_check_date`.
5. **Readiness check:** `python supervised_prep.py` must print "ready" before any training.

Item 1 (all bearings): defect types after dismantling

What you confirm:

Bearing	Defect type
B01	IR, OR, B
B02	IR
B03	IR
B04	IR, B

IR means inner race, OR outer race, B rolling element. The types are transcribed in `paderborn_experiments.json` and `label_table.json`.

Original documentation:

> Description p.8: "... the relevant threshold criterion for the end of an experiment and the defect type discovered after dismantling the test bearing."

> Description p.8, Table 3 "Data Set Characteristics". Rows: Experiment / Approx. Duration / No. of Files / Threshold Criterion / Defect. B01: 2 h, 377, "-", IR, OR, B. B02: 6 h, 1116, Tvibration, IR. B03: 3 h, 614, Ttemperature, IR. B04: 6 h, 1114, Tvibration, IR, B.

> Description p.8, under the table: "IR denotes inner race defect(s), OR denotes outer race defect(s) and B denotes rolling element defect(s)."

Our data: the file counts 377, 1116, 614 and 1114 equal our `.mat` counts and CSV rows (`checks.json`).

Assumptions and uncertainty:

- **Missing details:** Table 3 gives the type of damage only. It gives no size and no dismantling date.
- **Spelling:** "Tvibration" and "Ttemperature" are written as subscripted T in the PDF.

Recommendation: keep.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 2 (all bearings): assumptions A1–A4 and provisional end-stage labels

What you confirm: the four assumptions recorded in `label_table.json` are acceptable, with this reading of the deteriorated labels:

> **The final five measurements of B02, B03 and B04 are provisional end-stage proxy labels.** They stand for the end-of-test state of a bearing that was found damaged after dismantling (description Table 3). They are not a verified deteriorated state at those measurements. **The exact damage onset is unknown. The rig stopping at its vibration or temperature threshold does not by itself prove that the bearing defect caused the stop.**

- **A1, provisional end-stage proxy:** the last 5 measurements (about 1 minute) stand for the end-of-test state. That the damage found at dismantling was already present then is assumed, not verified.
- **A2, assumption:** the bearing defect caused the stop. The rig stopping at its threshold does not by itself prove this.
- **A3, not an onset time:** the exact damage onset is unknown. The deteriorated label marks the documented end stage, not when the damage began.
- **A4:** the bearing was healthy during its healthy-labelled rows.

Original documentation:

> Description p.2: "No defect was initiated in the bearings before the experiments."

> B02 log p.2, 08:59; B03 log p.2, 15:10; B04 log p.2, 12:45: "Begin with a new bearing."

> Description p.4: "A bearing fails and consequently an experiment ends, when at least one of two predetermined failure threshold criteria is exceeded."

What the documents do not say:

- when the bearing was dismantled;
- whether the rig ran again between the stop and the dismantling;
- the size of the damage;
- whether any inspection took place before the end.

Assumptions and uncertainty:

- **A4** rests on "no defect was initiated" and "new bearing".
- **A1 and A2** have no direct document. The description says a bearing "fails" when a criterion is exceeded, and Table 3 gives the damage found afterwards. Neither shows that the defect caused the stop, or that the damage existed in the final measurements.
- **A1 is the weakest assumption.** It covers only about 1.1–1.25 minutes before each stop.
- **A3** is a rule we impose on ourselves, so no document is needed.

Recommendation: keep the label ranges. Call the deteriorated labels "provisional end-stage proxy labels" in every result, with A1 and A2 named as the main uncertainties.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 3 (all bearings): label counts, and B01 withheld

What you confirm:

- **Counts:** 100 healthy rows at the start and 5 deteriorated rows at the end of each eligible bearing (`labels.py`, fixed 2 November before B02 and B04 were opened).
- **B01:** its healthy labels are withheld (review decision, 2 November; item 6).

Original documentation: none. These are design choices, not evidence. The 100 matches the anomaly baseline's commissioning period (about 30 minutes).

Our data:

Bearing	Healthy rows	Healthy period	Deteriorated rows
B02	M0052–M0151	09:09:44–09:39:51	5
B03	M0001–M0100	15:10:43–15:42:10	5
B04	M0051–M0150	12:55:27–13:26:01	5
B01	none	–	none

Assumptions and uncertainty:

- The counts are not evidence.
- The deteriorated-label count is to be checked for sensitivity at 3, 10 and 20 rows, within the development set only, and never used to choose a model.

Recommendation: keep.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 4 (B02, B04): end labels placed by a vibration stop rule

What you confirm: the B02 and B04 deteriorated labels are accepted. They sit where the rig's vibration limit fired, on the same rear accelerometer the model uses, under two conditions:

- every trained head is compared with that rule;
- B03, the temperature stop, is reported separately.

Original documentation:

> Description p.4: "The threshold criteria is based on the bearing housing vibration Tvibration and temperature Ttemperature measurements. For the provided experiments, Tvibration lies within [6, 10] g ..."

> Description p.8, Table 3: B02 "Tvibration", B04 "Tvibration".

> B02 log p.1: "Conversion Factor 10 g/V". p.2, 14:36: "The test was successfully completed because the RMS value of the accelerometer exceeded a value of 0.65 V four times."

> B04 log p.2, 18:26: "The test was successfully completed because the RMS values of an accelerometer exceeded the specified limit!"

> Description p.3: "Three one-directional accelerometers ... measure the vibration of the bearing via its housing. However, two (A and C) of the three accelerometer measurements ... are provided in this data set ..."

What the documents do not say:

- which accelerometer the stop rule watched;
- the RMS window it used;
- B04's limit value.

Our data:

- **B02's limit:** 0.65 V × 10 g/V = 6.5 g, inside the [6, 10] g range.
- **Rear sensor A in B02:** it is above 6.5 g at exactly the two measurements before the 13:32 pause (M0920, M0921).
- **Front sensor C:** never reaches 6.5 g in either bearing.
- **Last 5 rows above 6.5 g:** 4 of 5 in B02 (4.92, 7.16, 7.46, 6.57, 6.94 g) and 4 of 5 in B04 (3.17, 6.98, 8.72, 8.21, 6.81 g).
- **The rig rule as a detector** (rear RMS above 6.5 g, 3 in a row) catches 40% of the deteriorated rows on B02 and B04, and 0% on B03.

Assumptions and uncertainty:

- That the rule used sensor A is an inference from our data; no document says so.
- A network trained on these labels can do well simply by learning the stop rule.

Recommendation: keep, with both conditions.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 5 (B01): end label withheld

What you confirm: B01 gets no deteriorated label.

Original documentation:

> Description p.4: "However, for experiment B01 no criteria was pre-set …"

> Description p.8, Table 3: B01, Threshold Criterion "-", Defect "IR, OR, B".

> B01 log p.2, 09:13: "The experiment successfully completes because the temperature threshold value was exceeded!"

Our data:

- **Recording continues past the log:** 272 operating measurements were recorded after 09:13, up to 10:59. That is 106.5 minutes after the log's last entry, in two further segments.
- **Not covered by the rule:** our end-label rule requires the recording to end within 2 minutes of a documented stop, so B01 does not qualify.

Assumptions and uncertainty:

- The description and the log contradict each other about whether a criterion existed.
- What happened after 09:13, and when the bearing was dismantled, is undocumented.

Recommendation: keep it withheld.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 6 (B01): no healthy labels (decided 2 November)

What you confirm: B01 has no healthy labels, so it has no labels at all and is in no training fold.

Original documentation:

> B01 log p.2, 08:45: "The new bearing was installed the day before. It operated for a maximum of 15 minutes." Followed by the 10-minute warm-up.

> B01 log p.2, 09:13: "The experiment successfully completes because the temperature threshold value was exceeded!"

> Description p.4: "Ttemperature is 110 °C."

Our data:

- **Timing:** the warm-up runs 08:46:54–08:56:51 and operation starts at 08:57:10.
- **Temperature:** across the first 54 operating rows (M0052–M0105), T1 rises from 32.8 to 113.6 °C. It first reaches 110 °C at 09:12:49, about 16 minutes after warm-up.
- **Restarts:** the recording restarts at 09:32 and 09:43.
- **Vibration:** rear RMS stays at about 1.2 g (median).
- **Comparison:** B03 took about 3 hours to reach 110 °C; B02 and B04 peaked at 74 and 66 °C after 5–6 hours.

Assumptions and uncertainty:

- **Cause:** the documents don't explain the fast heating (lubrication, fit or preload).
- **Prior running:** "operated for a maximum of 15 minutes" means the bearing had run before this test.
- **Why the labels were dropped:** a bearing at the documented failure temperature is not a defensible example of a healthy start. Dropping B01 costs no deteriorated rows and 54 healthy rows.

Recommendation: keep the decision (B01 has no labels).

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 7 (B02): healthy labels M0052–M0151

What you confirm: B02's first 100 operating measurements, 09:09:44–09:39:51, are labelled healthy (0).

Original documentation:

> B02 log p.2, 25-10-2021 8:59: "Begin with a new bearing. Warm-up phase: 10 Minutes, n=750rpm, Fstatic = 2500N, no dynamic load. After the warm-up phase, the test runs with set values for rotation speed, static load and dynamic load."

> Description p.2: "No defect was initiated in the bearings before the experiments."

Our data:

- **Warm-up:** M0001–M0051 at 750 rpm and 2500 N, excluded from the labels.
- **Healthy rows:** about 30 minutes at test conditions, with rear RMS of about 1 g.
- **Temperature:** T1 is still rising, and the thermal calibration is valid.

Assumptions and uncertainty: A4 only. Nothing points against it.

Recommendation: keep.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 8 (B02): deteriorated labels M1112–M1116

What you confirm: B02's last 5 measurements, 14:35:29–14:36:36, are labelled deteriorated (1).

Original documentation:

> B02 log p.2, 25-10-2021 14:36: "The test was successfully completed because the RMS value of the accelerometer exceeded a value of 0.65 V four times."

> Description p.8, Table 3: B02, Tvibration, Defect "IR".

> Description p.4: "Tvibration lies within [6, 10] g."

Our data:

- **End of recording:** the last record is 36 s after 14:36, with no record after the logged stop.
- **Coverage:** the 5 rows span 1.1 minutes, all in segment 2 (after the 13:32 pause).
- **Vibration:** 4 of the 5 rows are above 6.5 g.

Assumptions and uncertainty:

- **A1, provisional end-stage proxy:** these rows stand for the end-of-test state; damage in them is assumed, not verified. A2: the stop does not by itself prove the defect caused it.
- **Item 4:** the position of these labels comes from the vibration rule.

Recommendation: keep.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 9 (B02): the 13:32 pause is not a label

What you confirm: the 13:32 stop is not used as a label or an onset time. The measurements around it stay unlabelled.

Original documentation:

> B02 log p.2, 25-10-2021 13:32: "The test was stopped because the RMS value of an accelerometer exceeded a value of 0.65V twice. To account for possible outliers, the test was continued, where the threshold is allowed to be crossed 4 times."

Our data:

- **Pause:** a gap from 13:32:36 to 13:37:18 (4 min 42 s).
- **Crossings:** exactly two rows above 6.5 g before it, M0920 and M0921.
- **Vibration level:** rear RMS rose from about 1 g to about 5 g at about 13:15.

Assumptions and uncertainty:

- The bearing was not dismantled at 13:32, so there is no evidence of its condition then.
- The rise at about 13:15 is a measurement, not a label.

Recommendation: keep it unlabelled.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 10 (B02): crossing count, 13 in our data and 4 in the log

What you confirm: the difference is acknowledged, and no label depends on it.

Original documentation:

> B02 log p.2, 14:36: "… exceeded a value of 0.65 V four times."

Our data: after the restart (M0922 onward), our 1.6-s RMS is above 6.5 g in 13 measurements.

Assumptions and uncertainty:

- **Not documented:** the rig's own RMS window, its counting rule, and whether it measured between our stored windows.
- **Most likely cause:** our RMS and the rig's monitoring RMS are computed differently.

Recommendation: acknowledge; no change.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 11 (B03): healthy labels M0001–M0100

What you confirm: B03's first 100 measurements, 15:10:43–15:42:10, are labelled healthy (0).

Original documentation:

> B03 log p.2, 05-11-2021 15:10: "Begin with a new bearing. Warm-up phase: 10 Minutes, n=750rpm, Fstatic = 2500N, no dynamic load. After the warm-up phase, the test runs with set values for rotation speed, static load and dynamic load."

> Description p.2: "No defect was initiated in the bearings before the experiments."

Our data:

- **No warm-up rows:** the first record is at 15:10:43, and no row is at 750 rpm. The warm-up the log describes is not in the recording.
- **Healthy rows:** about 31 minutes at test conditions, with rear RMS of about 1 g.
- **Temperature:** the thermal calibration is valid.

Assumptions and uncertainty: the missing warm-up is undocumented. It does not bear on whether the bearing was healthy.

Recommendation: keep.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 12 (B03): deteriorated labels M0610–M0614

What you confirm: B03's last 5 measurements, 18:22:25–18:23:40, are labelled deteriorated (1).

Original documentation:

> B03 log p.2, 05-11-2021 18:23: "The test was automatically stopped because the bearing housing temperature exceeded the specified limits of 110°C!"

> Description p.8, Table 3: B03, Ttemperature, Defect "IR". Description p.4: "Ttemperature is 110 °C."

Our data:

- **Temperature crossing:** T1 first goes above 110 °C at M0612 (18:23:02), inside the labelled rows. Three rows are above 110 °C.
- **End of recording:** the last record is 40 s after 18:23, at 118.2 °C.
- **Vibration:** rear RMS stays below 6.5 g in all 5 labelled rows (4.6–5.6 g).

Assumptions and uncertainty:

- **A1, provisional end-stage proxy:** these rows stand for the end-of-test state; damage in them is assumed, not verified. A2: the stop does not by itself prove the defect caused it.
- **Overshoot:** T1 reads 8 °C above the limit, probably from sensor lag or a different control sensor.
- **Why this bearing matters:** it is the only deteriorated label whose position was not chosen by vibration.

Recommendation: keep.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 13 (B04): healthy labels M0051–M0150

What you confirm: B04's first 100 operating measurements, 12:55:27–13:26:01, are labelled healthy (0).

Original documentation:

> B04 log p.2, 17-11-2021 12:45: "Begin with a new bearing. Warm-up phase: 10 Minutes, n=750rpm, Fstatic = 2500N, no dynamic load …"

> B04 log p.2, 13:45: "Since the bearing temperature was above 60°C, the parameters for the higher operating temperature were selected!"

> B04 log p.1: "… the bearing temperature is queried after 60 minutes of operation. If this exceeds 60°C, the set intervals of the parameters remain."

> Description p.2: "No defect was initiated in the bearings before the experiments."

Our data:

- **Warm-up:** M0001–M0050, excluded from the labels.
- **Healthy rows:** about 31 minutes, ending at 13:26, before the 13:45 parameter decision.
- **Set ranges:** they match the higher-temperature set (T_high) throughout (`checks.json`).
- **Temperature:** the thermal calibration is valid.

Assumptions and uncertainty: A4 only.

Recommendation: keep.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 14 (B04): deteriorated labels M1110–M1114

What you confirm: B04's last 5 measurements, 18:25:44–18:26:58, are labelled deteriorated (1).

Original documentation:

> B04 log p.2, 17-11-2021 18:26: "The test was successfully completed because the RMS values of an accelerometer exceeded the specified limit!"

> Description p.8, Table 3: B04, Tvibration, Defect "IR, B". Description p.4: "Tvibration lies within [6, 10] g."

Our data:

- **End of recording:** the last record is 58 s after 18:26.
- **Vibration:** 4 of the 5 rows are above 6.5 g, and the largest rear RMS in the whole run is 8.72 g.
- **Earlier alarm:** the anomaly baseline alarms 3.5 hours earlier (M0442), with unchanged operating ranges.

Assumptions and uncertainty:

- **A1, provisional end-stage proxy:** these rows stand for the end-of-test state; damage in them is assumed, not verified. A2: the stop does not by itself prove the defect caused it.
- **Limit value:** unknown. If our RMS matched the rig's, it would lie between 6 and 8.7 g.
- **Item 4:** the position of these labels comes from the vibration rule.
- **Not an onset:** the early alarm shows why the deteriorated label must not be read as the start of the damage.

Recommendation: keep.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	

Item 15 (B04): suspected logging or scaling discrepancy in speed and dynamic load

What you confirm: B04's speed and dynamic-load readings are documented as follows:

> **This is a suspected logging or scaling discrepancy, supported by our vibration analysis. It is not confirmed by the dataset authors.** In every B04 row, the recorded measured speed is 0.498–0.500 times the set speed, and the peak dynamic load is about 0.52 times the set value. The static load is normal. Our vibration analysis supports, but does not prove, that the shaft ran at the set speed and that these two columns were logged at half scale. Neither the dataset description nor the B04 log mentions it. The data are kept as recorded.

Original documentation:

> Description p.7: "meanAbs_speed / rpm is the mean absolute value of the measured shaft rotating speed." Also: "peak_dynLoad / N is the maximum value of the measured dynamic load".

> B04 log p.1: shaft rotational speed "Thigh [1500, 3000] rpm"; amplitude of the dynamic load "Thigh [500 1500] N".

Neither document mentions a scale factor, a fault or a correction.

Our data (the supporting evidence):

- **Ratios:** in all 1,114 rows, measured speed / set speed = 0.498–0.500 and peak dynamic load / set = 0.49–0.53 (median 0.52). B02 and B03 read 1.00 and about 1.06.
- **Spectrum:** in 14 vibration files spread from M0001 to M1114, the shaft-rotation line sits at the set speed, 40 to 326 times the background. At the recorded speed there is only background (0.1 to 2.7 times).
- **No effect on the model inputs:** they are identical with the speed doubled (a test checks this).
- **No effect on the labels:** labelling never reads the speed.

Assumptions and uncertainty:

- **Not confirmed:** half-scale logging is our interpretation of the spectrum, not a documented fact. Its cause (a logging scale factor, a sensor setting, or something else) is not documented and cannot be established from the files we have.
- **Recorded as a suspicion:** the registry's note for B04 (`paderborn_experiments.json`) now uses the same wording.
- **Future cross-bearing use:** raw speed or load must first multiply B04's two columns by 2, and it must say that the correction rests on this suspected discrepancy.

Recommendation: accept the documentation as a suspected discrepancy, keep the data as recorded, and optionally ask the authors (O. K. Aimiyekagbon, LDM, Paderborn University) to confirm.

	Confirm	Change	Comment
Reviewer	☐	☐	
Second checker	☐	☐	